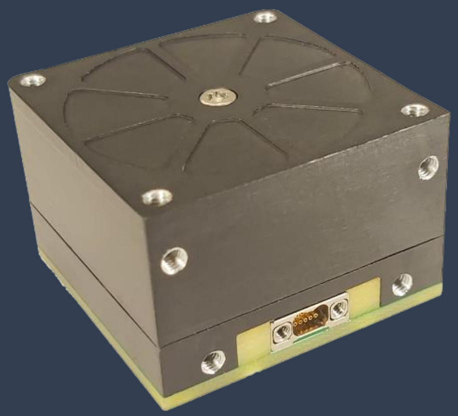
Cubesat Reaction Wheels (CubeWheel)

The reaction wheel is installed on the satellite and by combining the speed of reaction wheels, the satellite can be rotated around three axes or a combination of these three axes. Commands for each of the wheels installed on the satellite are sent by the command center to the satellite receiver and applied to the wheels via the internal network. The reaction wheel actuator is mainly used to control the status of the satellite. These actuators are able to provide the torque and momentum required to maneuver and stabilize the satellite.



Performance	Momentum	Nominal: 0.005 Nms @ 8500 rpm Peak: @10000 rpm (at 7 V supply)
	Max Torque	1 mNm at 0.005 Nms (at 5 V supply)
	Control mode	Speed or torque with built-in control CPU
	Command/telemetry	CAN, I2C(tbc)
	Reliability	99%
	Speed Accuracy	< 5 rpm
	Angular Rate feedback	12 bit/sec
	Temperature Sensor	On Wheels and Control Board
	Latch up protection	Constant current monitoring
	Control	Integrated electronics which includes drive circuitry and speed control algorithms
Electrical	Supply Voltage	Nominal: 6 to 10 V
	Power (Psystem + Pfriction + Pdynamic)	@10 V @ 1000RPM: < 0.3 W @ max speed: < 2.5 W
Environment	Thermal	-20 to +70° C
	Radiation	>10 krad dose
Mechanical	Dimensions	34 x 34 x 17 mm
	Mass	<80g